Where,

T_n = Net Barter Terms of Trade
P_x = Index number of export prices
P_m = Index number of import prices

This is used to measure the gain from international trade. If 'Tn' is greater than 100, then it is a favourable terms of trade which will mean that for a rupee of export, more of imports can be received by a country.

2 Gross Barter Terms of Trade

This was developed by Taussig in 1927 as an improvement over the net terms of trade. It is an index of relationship between total physical quantity of imports and the total physical quantity of exports.

$$T_g = (Q_x/Q_m) \times 100$$

Where, Q_m = Index of import quantities
Q_x = Index of export quantities

If for a given quantity of export, more quantity of import can be consumed by a country, then one can say that terms of trade are favourable.

3 Income Terms of Trade

The income terms of trade was given by G.S.Dorrance in 1948. It is the index of the value of exports divided by the price index for imports multiplied by quantity index of exports. In other words, it is the net barter terms of trade of a country multiplied by its exports-volume index.

$$T_y = (P_x / P_m)Q_x$$

Where, P_x = Price index of exports
P_m = Price index of imports
Q_x = Quantity index of exports

7.7.3. Terms of Trade related to the Interchange between Productive Resources

1 The Single Factoral Terms of Trade

Viner has devised another concept called "the single factoral terms of trade" as an improvement upon the commodity terms of trade. It represents the ratio of export-price index to the import-price index adjusted for changes in the productivity of a country's factors in the production of exports. Symbolically, it can be stated as

$$T_f = (P_x / P_m) F_x$$

Where, T_f stands for single factoral terms of trade index. F_x stands for productivity in exports (which is measured as the index of cost in terms of quantity of factors of production used per unit of export).

2 Double Factoral Terms of Trade

Viner constructed another index called "Double factoral terms of Trade". It is expressed as

$$T_{ff} = (P_x / P_m) \ (F_x / F_m)$$

which takes into account the productivity in country's exports, as well as the productivity of foreign factors.

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